Passport Authentication System

1) Introduction

1.1 Purpose

This document describes the requirements of a passport Authentication system. It explains the main functions used to verify passport details and confirm the identity of passport holders.

1.2 Scope

The system verifies passport information, applicant identity and document validity. It helps authorised officials detect invalid or suspicious passports.

1.3 Overview

The passport Authentication system is used to check whether a passport is genuine and belongs to the person presenting it. It verifies passport details against stored records and provides the authentication result.

2) General Description

The system is mainly used by immigration officers, government officials and authorised organisation. It provides features such as:

- Passport detail verification
- Identity verification
- Document validation
- Passport status checking
- Authentication reports

3) Functional Requirements.

3.1 Passport Verification

1) The system should accept passport details such as passpod neumber and DOB.
2) It should check the information against official records.
3) It should identify expired or invalid passpods.

3.2 Identity Verification

1) The system should compare the holder's details with passport records.
2) It may use photographs or biometric information for verification.
3) It should indicate whether the identity matches

3.3 Document Authentication

1) The system should verify security information available on the passpod.
2) It should detect altered or suspicious passpod details
3) The authentication result should be recorded.

3.4 Passport Status

1) The system should display whether a passpot is valid, expired, cancelled or reported lost.
2) Authorised users should be able to update passpot status.

3.5 Record Management
1) Maintain passport authentication history.
2) Store verification results securely
3) Allow authorised officials to search passport records

3.6 Reports
1) Generate authentication & verification reports.
2) Provide details of failed or suspicious verification attempts.

4) Non functional Requirements

4.1 Security
Passport and personal information must be protected from unauthorised access.

4.2 Performance
The system should verify passport details quickly

4.3 Reliability
Authentication results and passport records should be accurate and consistent.

4.4 Availability
The system should be available whenever passport verification is required

4.5 Scalability
The system should support a large number of passport records and verification requests.

5) Design constraints
- Only authorised officials should access passport records.
- The system must protect sensitive personal and biometric information
- It should follow government security and authentication standards.
- A secure network connections is required for accessing official records.

6) Hardware and Software requirements

Hardware : Computer/servers, passport scanner, biometric device and network connection

Software : OS, DBMS, authentication software application server and web browser.

7) Preliminary scheduling and requirement
The system may take around 10 weeks because of additional security and verification requirement. The estimated budget is approx. ₹80,000 covering development, security testing, authentication software and maintenance.

Activity	Duration
Requirement Analysis	1 week
System Design	2 weeks
Development	4 weeks
Testing & Security Testing	2 week
Deployment & Documentation	1 week
	10 weeks

Preliminary Budget

Resource	Estimated cost
Development	₹40,000
Security & Testing	₹15,000
Hardware / Scanning	₹10,000
Software / Tools	₹8000
Maintenance	₹7000
Total	₹80,000